SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS WITH ANSWERS

Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

1. Identify any incorrect or invalid fields in the header.

2. Analyze the issue with Content-Length and explain its impact.

3. Determine whether the status code matches the response correctness.

4. Suggest corrections for the identified errors.

Answers

1. The invalid field is `Content-Length: -150`. Content-Length cannot be negative. The other shown fields are generally valid in format.

2. Content-Length specifies the size of the response body in bytes. A negative value is invalid and can cause parsing errors or incorrect handling by clients and intermediaries.

3. `200 OK` indicates that the request succeeded. It does not make an invalid header field valid, so the response still needs correction.

4. Set Content-Length to the actual non-negative body size, for example `Content-Length: 150` if the body is exactly 150 bytes. Otherwise, use appropriate transfer framing and omit Content-Length when it should not be sent.

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: 150
Connection: keep-alive
```

2: DOM Structure Analysis

Consider the following DOM tree representation:

1. Identify missing or improperly closed elements.

2. Analyze how the browser will interpret this structure.

3. Determine the impact on rendering and layout.

4. Suggest corrections to make the DOM valid.

Answers

1. The <p> element is not closed before the element. The paragraph should be closed explicitly.

2. Browsers use HTML error-recovery rules to construct a DOM tree. They may automatically close the paragraph when encountering the closing div, but relying on this recovery is not good practice.

3. The page may still display, but unclear nesting can lead to unexpected spacing, layout differences, and harder maintenance.

4. Close the paragraph and keep the elements properly nested, as shown below.

```
<html>
  <head>
    <title>TestPage</title>
  </head>
  <body>
    <div>
      <p>HelloWorld</p>
    </div>
  </body>
</html>
```

3: GET vs POST in Login System

A login system uses two different methods:

1. Interpret the differences in data transmission between GET and POST.

2. Analyze which method is more suitable for login and why.

3. Identify security risks in using GET for authentication.

4. Suggest best practices for secure login implementation.

Answers

1. GET commonly sends parameters in the URL query string, where they can appear in browser history, logs, and copied links. POST sends submitted data in the request body, so credentials are not displayed in the URL.

2. POST is more suitable for login because it avoids placing usernames and passwords in the URL. However, POST alone does not encrypt data.

3. GET can expose credentials through browser history, server/proxy logs, bookmarks, screenshots, and referrer information. URLs may also be shared accidentally.

4. Use HTTPS, submit credentials using POST, store passwords securely with a strong password-hashing algorithm, validate inputs, use secure session cookies, apply rate limiting, and never log passwords. POST must be combined with HTTPS for confidentiality.

```
GET  /login?user=admin&pass=1234   Visible in URL   Low
POST /login                         Data in body    Better, but use HTTPS
```

4: HTML Code Inspection

Analyze the following HTML snippet:

1. Identify missing closing tags.

2. Analyze issues with the <img> tag.

3. Determine how browsers will handle these errors.

4. Suggest a corrected version of the code.

Answers

1. The `<h1>` and `<p>` elements are missing explicit closing tags. The image element is a void element and does not need a closing tag in HTML.

2. The `<img>` tag has a source but no `alt` text. Alternative text should be supplied for accessibility; also confirm that `image.jpg` exists at the specified path.

3. Browsers generally attempt to repair common HTML markup errors and display the page, but the result may not be exactly as intended. Missing alternative text also makes the image less accessible.

4. Use properly closed heading and paragraph tags and add descriptive alternative text.

```
<!DOCTYPE html>
<html>
<head>
  <title>Sample</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a paragraph.</p>
  <img src="image.jpg" alt="Sample image">
</body>
</html>
```

5: Browser-Server Communication Flow

The following sequence describes a web request: 1. User enters a URL in the browser; 2. DNS lookup occurs; 3. HTTP request is sent; 4. Server processes request; 5. Response is returned; 6. Browser renders page.

1. Interpret each step in the communication flow.

2. Identify where delays can occur.

3. Analyze the role of DNS in the process.

4. Suggest optimizations to improve performance.

Answers

1. The browser reads the URL; DNS resolves the domain name to an IP address; the browser connects to the server and sends an HTTP request; the server processes it; the server returns an HTTP response; and the browser parses the response and renders the page.

2. Delays can occur during DNS resolution, connection setup, TLS negotiation for HTTPS, network transmission, server-side processing, database access, downloading resources, and page rendering.

3. DNS translates a human-readable domain name, such as `example.com`, into an IP address that the browser can use to contact the server. DNS caching can reduce repeated lookup time.

4. Use caching, a CDN, compressed files, optimized images, fewer unnecessary requests, efficient server/database code, persistent connections, and suitable DNS caching. Use HTTPS and optimize critical resources without weakening security.

CODE OF CONDUCT

I _____ (Name / Reg No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misreads or fails to understand data.
Analysis	25%	Deep analysis with strong logical reasoning and justification	Reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided.
Application of Concepts	20%	Effectively applies relevant concepts to the problem	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data.
Conclusion	20%	Clear, meaningful conclusions with strong insights	Valid conclusions with adequate insights	Conclusions are vague or weak	No clear or valid conclusions.
Clarity	10%	Clear presentation using proper terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand